# LECTURE 8

# INSTRUMENTAL VARIABLES ESTIMATION II

(A) Two-Stage Least Squares

(B) Checking Instrument Validity: Instrumental Relevance

(C) Checking Instrument Validity: Instrumental Exogeneity

(D) Testing for Endogeneity: Hausman test

## (A) Two-Stage Least Squares

- When more instruments than endogenous explanatory variables are available (overidentified case), how do we run the IV estimation?

- A multiple regression model with one endogenous explanatory variables

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u,$$

where $x_1$ is endogenous and $x_2$ is exogenous.

Suppose we have more than one instrumental variable. $z_1$ and $z_2$ are two instruments for endogenous $x_1$. Which one do we use for $x_1$?

- We should consider using both instruments.
- But how? Choose a linear combination of both instruments.

- (In the first -stage estimation) we run regression of

$$x_1 \text{ on } z_1 \; z_2 \text{ and } x_2 \qquad (1)$$

Its fitted value is a linear combination of $z_1$, $z_2$ and $x_2$.

Since exogenous $x_2$ could be correlated with $x_1$, we include $x_2$ in the regression, i.e., we run the following regression model.

$$x_1 = \pi_0 + \pi_1 z_1 + \pi_2 z_2 + \pi_3 x_2 + v$$

The estimated regression model is

$$\hat{x}_1 = \hat{\pi}_0 + \hat{\pi}_1 z_1 + \hat{\pi}_2 z_2 + \hat{\pi}_3 x_2.$$

This is an auxiliary regression, so coefficients $\hat{\pi}_1, \hat{\pi}_2, \hat{\pi}_3$ here have no behavioral interpretations.

Since $\hat{x}_1$ is a linear combination of $z_1$, $z_2$ and $x_2$, $Cov(\hat{x}_1, u) = 0$. Now $\hat{x}_1$ can be regarded as an instrument for $x_1$.

- (In the ___second___-stage estimation) we replace $x_1$ with $\hat{x}_1$ in the regression

$$y \text{ on } \hat{x}_1 \text{ and } x_2 \qquad (2)$$

The resulting OLS are the 2-Stage Least Squares (2SLS) estimators of $\hat{\beta}_{0,2SLS}, \hat{\beta}_{1,2SLS}, \hat{\beta}_{2,2SLS}$.

- (1) is called the first-stage regression, and (2) is called the second-stage regression.

- In Stata, it combines (1) and (2) regressions together and just run one regression:

(suppose mother's education (*meduc*) is another instrument for *educ*)

Example: Return to Education

Stata> ivreg lwage (educ = sibs meduc) married

$$\ln(\text{wage}) = \beta_0 + \beta_1 educ + \beta_2 married + u$$

$educ$ is endogenous, and the instrumental variables are $sibs(z_1)$ and $meduc(z_2)$.

```
. ivreg lwage (educ=sibs meduc) married

Instrumental variables (2SLS) regression

      Source |       SS           df       MS      Number of obs   =       857
-------------+----------------------------------   F(2, 854)       =     34.99
       Model |  6.83422944         2  3.41711472   Prob > F        =    0.0000
    Residual |  142.526811       854  .166893221   R-squared       =    0.0458
-------------+----------------------------------   Adj R-squared   =    0.0435
       Total |   149.36104       856  .174487197   Root MSE        =    .40853

------------------------------------------------------------------------------
       lwage |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        educ |   .1164975   .0163514     7.12   0.000     .0844039    .1485911
     married |    .243131   .0455303     5.34   0.000     .1537667    .3324953
       _cons |   4.994835   .2316247    21.56   0.000     4.540214    5.449455
------------------------------------------------------------------------------
Instrumented:  educ
Instruments:   married sibs meduc
------------------------------------------------------------------------------
```

- How does Stata/we get these coefficient results?

Original Model:

$$\ln(\text{wage}) = \beta_0 + \beta_1 educ + \beta_2 married + u$$

<u>In the first stage</u>, the model is

$$educ = \pi_0 + \pi_1 sibs + \pi_2 meduc + \pi_3 married + v,$$

and we get the estimated model and the fitted values $\widehat{educ}$.

$$\widehat{educ} = \hat{\pi}_0 + \hat{\pi}_1 sibs + \hat{\pi}_2 meduc + \hat{\pi}_3 married$$

<u>In the second stage</u>, we replace the *educ* by <u>educ_hat</u>

$$\ln(\text{wage}) = \beta_0 + \beta_1 \widehat{educ} + \beta_2 married + u$$

With first-stage regression result:

```
. ivregress 2sls lwage (educ = sibs meduc) married,first

First-stage regressions
-----------------------

                                                Number of obs   =        857
                                                F(   3,    853) =      51.78
                                                Prob > F        =     0.0000
                                                R-squared       =     0.1540
                                                Adj R-squared   =     0.1511
                                                Root MSE        =     2.0247

------------------------------------------------------------------------------
        educ |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
     married |  -.3345786   .2234766    -1.50   0.135     -.773207    .1040498
        sibs |  -.1404924   .0319393    -4.40   0.000    -.2031811   -.0778036
       meduc |   .2480149   .0253574     9.78   0.000     .1982447    .2977851
       _cons |   11.62972   .3785686    30.72   0.000     10.88669    12.37276
------------------------------------------------------------------------------


Instrumental variables (2SLS) regression               Number of obs =     857
                                                       Wald chi2(2)  =   70.23
                                                       Prob > chi2   =  0.0000
                                                       R-squared     =  0.0458
                                                       Root MSE      =  .40781

------------------------------------------------------------------------------
       lwage |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        educ |   .1164975   .0163227     7.14   0.000     .0845056    .1484895
     married |    .243131   .0454505     5.35   0.000     .1540497    .3322124
       _cons |   4.994835   .2312189    21.60   0.000     4.541654    5.448016
------------------------------------------------------------------------------
Instrumented:  educ
Instruments:   married sibs meduc
```

## Remarks:

(a) The IV estimation is a special case of the 2SLS estimation when the number of instruments is equal to the number of endogenous explanatory variable (exactly identified case).

(b) The 2SLS estimator is basically an IV estimator in nature. Compared with OLS with endogeneity, the 2SLS estimator is consistent with large variance.

## (B) Checking Instrument Validity: Instrumental Relevance

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$$

$y$: dependent variable (e.g., ln(wage)), $x_1$ is endogenous (e.g., *educ*), $x_2$ is exogenous (e.g., married). $z$ is an instrument for endogenous $x_1$.

Recall:

Two conditions for a valid instrument $z$:

(1) Instrument Exogeneity: $Cov(z, u) = 0$

(2) Instrument Relevance: $Cov(z, x) \neq 0$

- For the condition (1), we are not able to directly verify $Cov(z, u) = 0$ because the $u$ is unobservable. Usually, we use economic intuition to justify this.
- However, for the over-identified cases (($m$) # of instruments > ($p$) # of endogenous explanatory variables), we can statistically test for the (1) instrument exogeneity condition. See Section (C).

- Before running the IV (or 2SLS) estimation, we need to check for the instrument relevance, $Cov(z, x) \neq 0$. A simple *t*-test is used to test for the slope coefficients of the instrumental variable $z$ (in the first-stage).

Example: Return to Education with two explanatory variables.

$$\log(\text{wage}) = \beta_0 + \beta_1 educ + \beta_2 married + \varepsilon$$

$educ$ is endogenous; instruments: *sibs*.

The first-stage regression's result.

```
. reg educ sibs married

      Source |       SS       df       MS              Number of obs =     935
-------------+------------------------------           F(  2,   932) =   30.17
       Model |  274.065107     2  137.032554           Prob > F      =  0.0000
    Residual |  4232.75414   932   4.5415817           R-squared     =  0.0608
-------------+------------------------------           Adj R-squared =  0.0588
       Total |  4506.81925   934  4.82528828           Root MSE      =  2.1311

------------------------------------------------------------------------------
        educ |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        sibs |  -.2281621   .0302362    -7.55   0.000     -.287501   -.1688232
     married |  -.4234119   .2255123    -1.88   0.061    -.8659826    .0191588
       _cons |   14.51764   .2312577    62.78   0.000      14.0638    14.97149
------------------------------------------------------------------------------
```

*sibs* is statistically and economically significant in this regression.

- When the instrumental variable *z* is not significant in the first-stage regression, it is called weak ______ instrument.

- If *z* and *x* are weakly correlated, then the IV estimator has a large standard error.

- A rule of thumb for checking for two or more weak instruments: Consider one endogenous explanatory variable case. If the *F*-statistic for the joint significance of instruments in the first-stage regression estimation is less than 10___, then the instruments are weak.

Example: Return to Education (with two instruments: *sibs*, *meduc*)

In the first-stage:

$$educ = \pi_0 + \pi_1 sibs + \pi_2 meduc + \pi_3 married + v$$

The *F* test is used to test for the joint significance of instruments.

$$H_0\!: \pi_1 = 0, \pi_2 = 0 \quad \text{against } H_1\!: H_0 \text{ is wrong}$$

```
. reg  educ sibs meduc  married

      Source |       SS       df       MS              Number of obs =     857
-------------+------------------------------           F(  3,   853) =   51.78
       Model |  636.737512     3  212.245837           Prob > F      =  0.0000
    Residual |  3496.65805   853  4.09924743           R-squared     =  0.1540
-------------+------------------------------           Adj R-squared =  0.1511
       Total |  4133.39557   856  4.82873314           Root MSE      =  2.0247

------------------------------------------------------------------------------
        educ |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        sibs |  -.1404924   .0319393    -4.40   0.000    -.2031811   -.0778036
       meduc |   .2480149   .0253574     9.78   0.000     .1982447    .2977851
     married |  -.3345786   .2234766    -1.50   0.135     -.773207    .1040498
       _cons |   11.62972   .3785686    30.72   0.000     10.88669    12.37276
------------------------------------------------------------------------------

. test  sibs meduc

 ( 1)  sibs = 0
 ( 2)  meduc = 0

       F(  2,   853) =   76.14
            Prob > F =    0.0000
```

The *F*-statistic is 76.14 >>10. So, *sibs* and *meduc* are strong instrumental variables for *educ*.

## (C) Checking Instrument Validity: Instrumental Exogeneity

- When the (1) instrument exogeneity $Cov(z, u) = 0$ condition fails, the instrument $z$ is invalid. In this case, the IV estimator is inconsistent and misleading.

- For the exactly identified case (($m$) # of instruments = ($p$) # of endogenous explanatory variable), it is impossible to test for the (1) instrument exogeneity.

- In the overidentified case ($m > p$), e.g.,

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$$

$x_1$ is endogenous and $x_2$ is exogenous, and $z_1, z_2$ are two instruments for $x_1$.

We can test whether additional instruments are uncorrelated with $u$. For example, ***given*** that $z_1$ is a __valid__ instrument, we can test if $z_2$ satisfies the (1) instrument exogenity. We ask: Is $Cov(z_2, u) = 0$?

- Idea: Under the null hypothesis $Cov(z, u) = 0$, the IV estimators $\hat{\beta}_{0,IV}$, $\hat{\beta}_{1,IV}$, $\hat{\beta}_{2,IV}$ are consistent, so is the IV residual $\hat{u}_{IV} = y - \hat{\beta}_{0,IV} - \hat{\beta}_{1,IV}x_1 - \hat{\beta}_{2,IV}x_2$.

- $u$ becomes "observable" with the residual $\hat{u}_{IV}$.

- $\hat{u}_{IV}$ is expected to be *approximately* uncorrelated with $z_1, z_2$ (as $\hat{u}_{IV}$ is based on sample).

- To see if it is the case, we regress $\hat{u}_{IV}$ on $z_1, z_2$ (and $x_2$) and test the joint significance of $z_1, z_2$, and then calculate the (*J* or Sargan) test statistic based on the *F* statistic for the joint significance, i.e., $J = mF$, where *m* is # of instruments. This is called the "test of overidentification restrictions".

- Under the null hypothesis that all the instruments are exogeneous, *J* has a Chi-squared (*m*–*p*) distribution. If *m* = *p*, *J* = 0, this test is invalid.

- The *J or* Sargan test

$$\hat{u}_{IV} = \gamma_0 + \gamma_1 z_1 + \gamma_2 z_2 + \gamma_3 x_2$$

$$H_0\text{: z}_2 \text{ is exogenous} \quad H_1\text{: z}_2 \text{ is not exogenous}$$

$$J = mF \sim x^2(m - p)$$

If some instrumental variables are exogenous and others are endogenous, the *J* statistic will be large, and the null hypothesis will be rejected.

## (D) Testing for Endogeneity: Hausman test

- We could also use the Hausman test to (indirectly) test for endogeneity. It is another way to check the valid of IV estimator. Instead of looking at instruments, we check whether we should trust OLS.

$$H_0\!: Cov(x_1, u) = 0 \text{ vs. } H_1\!: Cov(x_1, u) \neq 0$$

If we reject the null hypothesis, that means the model suffers Endogeneity ___________.

- Conceptually, we can compare the 2SLS and OLS estimates.

- Under the null hypothesis, both 2SLS and OLS estimates are consistent. So, the difference between $\hat{\beta}_{2SLS}$ and $\hat{\beta}_{OLS}$ should be small. Otherwise, we reject the null hypothesis.

$$H_0\!: \hat{\beta}_{2SLS} \; and \; \hat{\beta}_{OLS} \; are \; both \; consistent \;\; \text{against} \; H_1\!: only \; \hat{\beta}_{2SLS} \; consistent$$

Example: Return to Education with two explanatory variables

$$\log(\text{wage}) = \beta_0 + \beta_1 educ + married + u$$

$educ$ is endogenous; instruments: *sibs*, *meduc*

The Hausman test procedure is as follows.

Step 1: run OLS and store the OLS estimates in <u>name 1</u>

Step 2: run 2SLS using instruments and store the 2SLS estimates in <u>name 2</u>

Step 3: run hausman test: hausman <u>name 2</u> <u>name 1</u>

# Step 1: run OLS and store OLS estimates in *myOLS*

```
. reg lwage educ married

      Source |       SS           df       MS      Number of obs   =       935
-------------+----------------------------------   F(2, 932)       =     67.07
       Model |  20.8435506         2  10.4217753   Prob > F        =    0.0000
    Residual |  144.812733       932  .155378469   R-squared       =    0.1258
-------------+----------------------------------   Adj R-squared   =    0.1239
       Total |  165.656283       934  .177362188   Root MSE        =    .39418

------------------------------------------------------------------------------
       lwage |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        educ |   .0617349   .0058817    10.50   0.000     .0501919    .0732779
     married |   .2299469   .0417834     5.50   0.000     .1479464    .3119474
       _cons |   5.742176    .090445    63.49   0.000     5.564677    5.919676
------------------------------------------------------------------------------
```

# Step 2: run 2SLS using instruments and store estimates in *my2SLS*

```
. ivreg lwage (educ =sibs meduc) married

Instrumental variables (2SLS) regression

      Source |       SS           df       MS      Number of obs   =       857
-------------+----------------------------------   F(2, 854)       =     34.99
       Model |  6.83422944         2  3.41711472   Prob > F        =    0.0000
    Residual |  142.526811       854  .166893221   R-squared       =    0.0458
-------------+----------------------------------   Adj R-squared   =    0.0435
       Total |   149.36104       856  .174487197   Root MSE        =    .40853

------------------------------------------------------------------------------
       lwage |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        educ |   .1164975   .0163514     7.12   0.000     .0844039    .1485911
     married |    .243131   .0455303     5.34   0.000     .1537667    .3324953
       _cons |   4.994835   .2316247    21.56   0.000     4.540214    5.449455
------------------------------------------------------------------------------
Instrumented:  educ
Instruments:   married sibs meduc
------------------------------------------------------------------------------
```

# Step 3: run Hausman test

```
. hausman my2SLS myOLS

                 ---- Coefficients ----
             |      (b)          (B)            (b-B)     sqrt(diag(V_b-V_B))
             |     my2SLS       myOLS        Difference          S.E.
-------------+----------------------------------------------------------------
        educ |    .1164975     .0617349        .0547626        .0152569
     married |     .243131     .2299469        .0131841        .0180873
------------------------------------------------------------------------------
                           b = consistent under Ho and Ha; obtained from ivreg
            B = inconsistent under Ha, efficient under Ho; obtained from regress

    Test:  Ho:  difference in coefficients not systematic

                  chi2(2) = (b-B)'[(V_b-V_B)^(-1)](b-B)
                          =       13.10
                Prob>chi2 =      0.0014
```

The statistic 13.16 > 5.99, the critical value of Chi-squared (2) at 5% significance; the *p*-value (0.0014) is much smaller than 0.05. We reject the null hypothesis that the x is exogenous. We suspect *educ* is endogeneous. Thus, IV is favored.

Alternatively, in Stata, you can obtain Hausman test directly after 2SLS using command *estat endogenous*

```
. quietly ivregress 2sls lwage (educ= sibs meduc) married

. estat endogenous

  Tests of endogeneity
  Ho: variables are exogenous

  Durbin (score) chi2(1)          =  14.3563  (p = 0.0002)
  Wu-Hausman F(1,853)             =  14.5327  (p = 0.0001)
```

Similar result can be obtain. (same idea but different test statistics used).